UI Design Documentation including Evaluation Design

Booking Travel Mobile App

1. Overall Interface Layout

The Booking Travel Mobile App is positioned as a one-stop mobile travel planning tool. It does not try to become a full commercial booking and payment platform. Its interface therefore concentrates on five user goals: discover destinations, compare suggestions, create an itinerary, understand places on a map, and manage saved trips.

The design follows a bottom-navigation mobile structure. The five primary destinations are Home, Explore, Itinerary, Saved, and Profile. This keeps high-frequency functions within one tap and reduces the gulf of execution for first-time users.

Design principle	Application in the app
Visibility and feedback	Primary actions, loading states, saved states, validation errors, and map failures are visible instead of hidden.
Consistency	**Material Design 3** components and shared color roles are reused across cards, chips, buttons, dialogs, and bottom sheets.
Error prevention and recovery	Input validation, confirmation dialogs, undo/snackbar feedback, and retry actions reduce destructive or confusing outcomes.
Reduced memory load	Search history, recent trips, timeline grouping, and visible filters reduce the need for users to remember previous steps.
User control	Users can save, edit, reorder, delete, duplicate, and return from core flows without being locked into a wizard-like path.

1.1 Home Screen / Dashboard

The Home Screen is the main entry point for destination discovery, personalized recommendations, quick itinerary access, and map exploration. It uses a clean **mobile-first layout** with a top app bar, prominent search bar, recommendation cards, category filter chips, an upcoming-trip widget, and bottom navigation.

Note: Material Design 3 supports adaptive layouts across different screen types and sizes. Therefore, the Home Screen can maintain a consistent and usable experience on standard smartphones, foldable devices, and tablets.

- **Top area:** app title, avatar/profile shortcut, notification icon, and search field with placeholder text such as 'Search city, attraction, or travel theme'.
- **Main content:** 'Recommended for You' cards generated from explicit preferences, saved content, browsing history, or simple rule-based matching.
- **Quick actions:** Create New Itinerary, View Upcoming Trip, and Explore Map are placed near the top because they represent the highest-value tasks.
- **States:** first launch, returning user, empty recommendations, no network, and loading states all have explicit text and next actions.

1.2 Itinerary Management Screen (Core UI)

The Itinerary Management Screen is the core working interface. It **organizes travel dates, destinations, attractions, activities, notes, and editable plan items in a chronological timeline.** It supports create, edit, delete, reorder, and save operations from Project1.

- A vertical **timeline** groups items by day and keeps time, place, note, and status visible.
- Each item **card** includes edit, delete, move, open map, and duplicate actions.

- A floating action button (**FAB**) opens an add-activity bottom sheet so the user stays in planning context.
- The screen provides saved, unsaved, offline, and validation-error feedback.

1.3 Interactive Map Visualization

The **map** supports spatial understanding during planning and travel. It is not a replacement for a full navigation product. Its role is to show attraction distribution, route context, and the relationship between saved places and the itinerary.

- Markers are categorized by destination, attraction, restaurant, transport, saved place, and itinerary item.
- Tapping a **marker** opens a bottom sheet with place name, distance, category, short description, and Add to Itinerary.
- The map supports zoom, drag, marker filtering, and switching between exploration and planned-itinerary views.
- If map loading fails, a retry button and readable explanation replace any blank state.

1.4 Travel History & Saved Trips

Saved Trips and **Travel History** centralize current, future, and past travel materials. Upcoming trips are displayed first, followed by saved destinations, wishlists, and past trips.

- Saved trip cards show destination, date range, number of planned items, and last updated time.
- Wishlists can be grouped into folders such as Weekend, Family, Food, Nature, and City Break.
- Past trips can be searched or filtered by date and destination; duplicating a past trip is an optional future enhancement.

1.5 Profile, Settings & Customization

Profile and **Settings** support account access, phone verification login, social login, preference configuration, notification control, language/currency settings, privacy options, help, and visual theme customization.

- Account setup includes phone number verification and third-party social login.
- Travel preference tags include adventure, relaxation, food, culture, shopping, family, nature, and budget level.
- Theme settings use Material You / Material 3 color roles, not isolated manually selected colors.
- Privacy settings clarify what travel data is saved and which parts are local or account-related.

1.6 The Calling Relationship of Interface Functions (UML)

The interface calling relationship follows the Project1 functional modules: authentication, destination discovery, personalized recommendation, destination detail viewing, itinerary planning, interactive map, saved trip management, and user support. UML Diagram and table are shown. **More UML Diagram are shown in Appendix A, including Class Diagram, Component Diagram, Deployment Diagram and Package Diagram.**

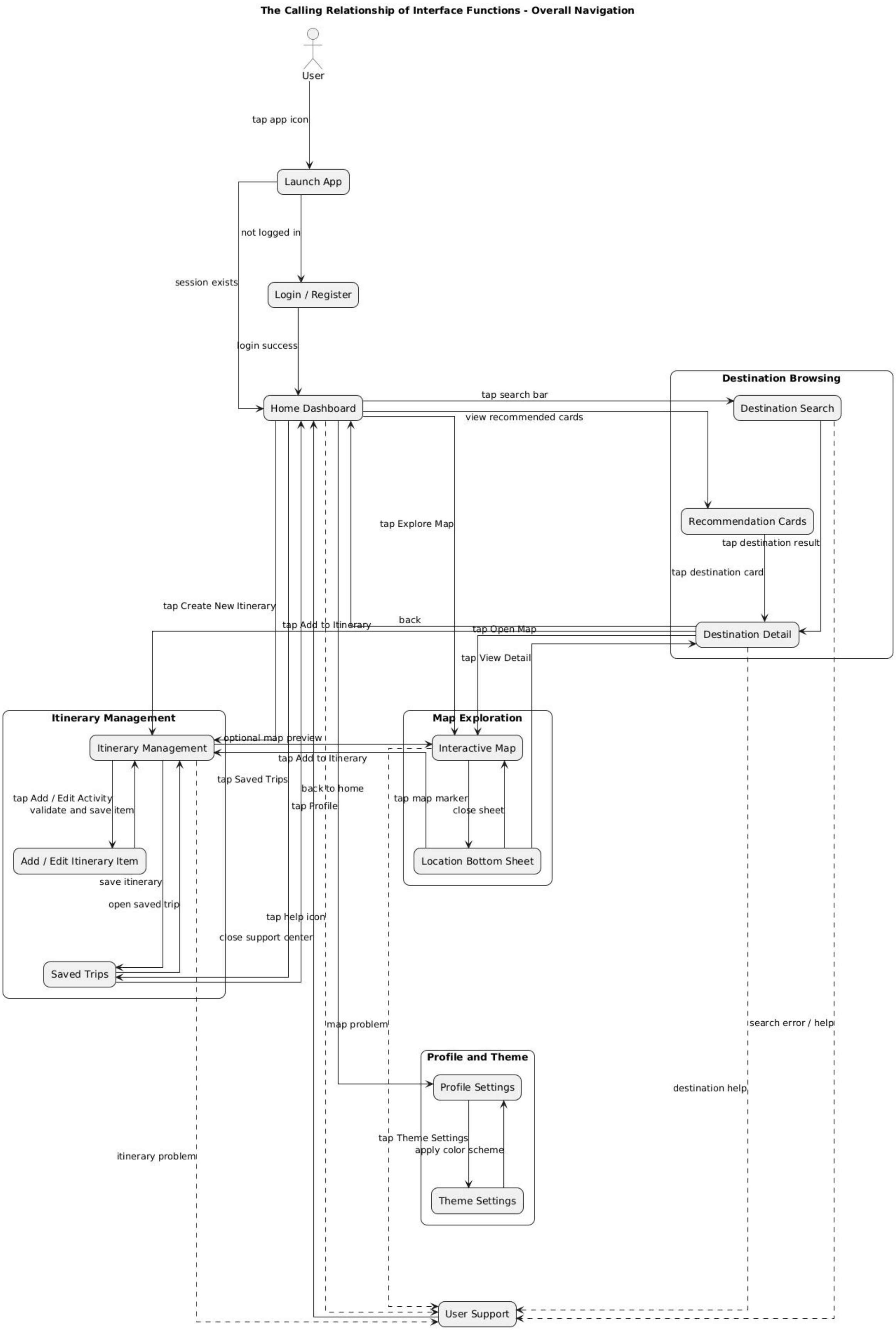


Figure 1. Overall navigation and interface calling relationship.

Interface	Triggered by	Calls / opens	Main output
Login / Register	Phone number, SMS code, social login button	Authentication service and session manager	Login status and entry to Home
Home Dashboard	App launch or Home tab	Recommendation list, search module, saved trip shortcut	Cards, chips, quick actions
Destination Search	Search bar and filter chips	Destination service and result list	Matched destinations and attractions
Destination Detail	Destination card	Detail service, map shortcut, itinerary action	Destination information and action buttons
Itinerary Management	Create/Edit itinerary action	Itinerary editor and validation module	Chronological trip plan
Interactive Map	Explore Map or Open Map action	Map service, marker renderer, location sheet	Map markers and location details
Saved Trips	Saved tab or upcoming trip card	Trip storage and history list	Saved itineraries and folders
User Support	Help icon, error prompt, Profile support entry	FAQ, onboarding, feedback, contact resources	Guidance and troubleshooting

1.7 User-Centered Analysis: Personas, Scenarios, and HTA (Hierarchical Task Analysis)

This part is newly added to the document. Its purpose is to strengthen the connection between user needs and interface design decisions. When developing an app, the design process should start from understanding who the users are, what goals they want to achieve, and what functions they are likely to use. Therefore, instead of directly "drawing" the user interface, this section first analyzes the users, usage scenarios, and task structure.

The lecture material emphasizes knowing the user, using personas, writing scenarios, and decomposing tasks. Based on this idea, the following analysis translates the target users described in the SRS into concrete interface design checks.

Persona	Needs	Design implications
University weekend traveler	Fast trip ideas, low learning cost, easy saving before holidays.	Home recommendations, quick filters, lightweight onboarding, and one-tap itinerary creation.
Independent traveler	Flexible plan adjustment, location awareness, and saved information during travel.	Editable timeline, map marker bottom sheet, offline/retry states, and quick access to current trip.
Occasional family planner	Clear structure, reduced risk of losing plans, and help when stuck.	Confirmation dialogs, visible saved state, help center, larger touch targets, and readable typography.

Core scenario used for evaluation:

- A student wants to plan a two-day weekend trip. They open the app, search a city, review recommendations, save one destination, create an itinerary, add an attraction from the map, edit the activity time, and check the saved trip before departure.
- A traveler is already outside and needs to quickly find the next planned attraction. They open the current trip, tap Open Map, read the bottom sheet, and adjust the item if the attraction is too far away.

HTA step	Subtasks	UI support
0. Plan a short trip	Choose destination, create trip, add activities, review map, save plan	Home, Search, Itinerary, Map, Saved
1. Choose destination	Enter keyword, select filters, compare cards	Search bar, chips, recommendation cards
2. Create itinerary	Set dates, name trip, add first item	Create button, date picker, bottom sheet
3. Add place from map	Open map, select marker, read details, add to	Map markers, bottom sheet, Add to

	itinerary	Itinerary
4. Adjust plan	Edit time, reorder item, delete mistake, save	Timeline card actions, confirmation, snackbar
5. Reopen during travel	Open current trip, locate item, access support if needed	Upcoming trip widget, Saved tab, Help Center

2. Home Screen / Dashboard

2.1 Personalized Recommendations

Recommendations are presented as helpful suggestions rather than opaque algorithmic decisions. The current project scope supports interpretable logic based on preferences, saved content, and history.

- Recommendation cards show image, destination name, theme, estimated duration, and reason such as 'matches your nature preference'.
- Primary card action is View Detail; secondary actions are Save and Add to Itinerary.
- Recommendation reasons are short and understandable so users can evaluate why an item appears.
- If no preference data exists, the screen switches to popular categories and asks users to select interests.

2.2 Destination Discovery & Search

Destination discovery combines a visible search bar, filter chips, and results cards. It supports destination, attraction, and travel-theme queries.

- Search bar remains near the top of Home and Explore so the user's next action is visible.
- Filter chips include Weekend Trip, City Break, Nature, Food, Culture, Family, and Budget Friendly.
- No-result state explains that no destination matched and offers to clear filters or try a broader keyword.
- Search validation handles empty input and unsupported characters with inline messages.

2.3 Quick Actions

Quick actions reduce the number of steps needed to reach the major tasks. They are placed above secondary content because they directly support the product objectives from Project1.

- **Create New Itinerary** opens trip creation.
- **View Upcoming Trip** opens the nearest saved trip.
- **Explore Map** switches to the map view with filters and destination markers.
- **Help** / How to Start opens short onboarding for first-time users.

2.4 Design Critique Rationale

Critique dimension	Assessment	Design response
First impression	The Home Screen must immediately communicate discovery and planning.	Search, recommendations, and current trip widget occupy the first viewport.
Usability	Too many travel options can create decision overload.	Use category chips, limited recommendation cards, and clear primary actions.
Visual hierarchy	The main CTA should not compete with all cards equally.	Create New Itinerary uses filled button/FAB styling; secondary actions are lower emphasis.
Consistency	Search, cards, and filter chips must behave the same across Home and Explore.	Reuse Material 3 component patterns and naming.
Accessibility	Outdoor travel contexts make low contrast and small targets risky.	Use high-contrast text, 44x44 touch targets, and scalable body text.

3. Itinerary Management Screen (Core UI)

3.1 Integrated Timeline View

The timeline view is the primary structure for itinerary planning. It shows the plan chronologically and supports quick adjustment when real travel conditions change.

- **Each day** has a section header with date, city, and optional weather/reminder placeholder.
- **Each itinerary** item is a card with time, attraction/activity, location, note, and an action menu.
- **Drag** handle or **Move** action supports reordering within a day or moving to another day.

3.2 Key Information

The itinerary page shows planning-relevant information and avoids implying that the product handles actual hotel, flight, or payment transactions.

- Required fields: date and at least one itinerary content field, such as place, attraction, or activity.
- Optional fields: notes, time range, tags, and map location reference.
- System status: saved, unsaved changes, offline/network unavailable, and validation errors.

3.3 Immediate Actions

- **Add** Activity / Attraction: primary floating action button.
- **Edit** Item: opens modal bottom sheet with pre-filled fields.
- **Delete** Item: opens confirmation dialog and offers undo where feasible.
- Open **Map**: shows the selected place in the map view.
- **Save** Plan: stores the latest itinerary state and updates Saved Trips.

3.4 Error Prevention and Recovery

Risk	Preventive UI	Recovery UI
Missing date or empty activity	Inline validation before save	Explain required field and keep entered content
Accidental deletion	Confirmation dialog for destructive delete	Snackbar undo if technically feasible
Unsaved changes	Visible unsaved state and save prompt	Keep draft locally until user decides
Network/save failure	Progress state and disabled repeated submit	Retry, save draft, and readable failure message

4. Interactive Map Visualization

4.1 Explore Mode

Explore Mode is a full-screen map interface for browsing destinations and attractions. It provides spatial context without replacing specialized navigation software.

- **Top app bar** includes back button, search entry, and filter icon.
- **Markers** are clustered or filtered when too many items are visible.
- **Bottom navigation** can be hidden while the map is in focused exploration mode, then restored when the user exits.

4.2 Location-Based Details

Location details are shown in a modal bottom sheet to keep map context visible while presenting actionable information.

- Bottom sheet includes place name, category, distance, short description, and relevant actions.
- Primary action is Add to Itinerary; secondary actions include Save Place and View Detail.
- Map permission, loading, and unavailable-data states are explicit and recoverable.

4.3 Map Accessibility and Performance

- Map pins need text labels or list equivalents so users are not forced to depend on visual markers only.
- Marker tap areas should meet mobile target-size expectations, especially in outdoor use.
- Map data loading should show progress and avoid blocking access to already saved itinerary information.

5. Travel History & Saved Trips

5.1 Past Trips (Travel History)

Past Trips display completed or older travel plans. This helps users review previous itineraries and reuse useful patterns while staying within the course-project scope.

- **Past trip** cards include destination, date range, number of items, and optional note summary.
- **History** can be searched by destination or filtered by year/month.
- **Duplicating** a past trip into a new itinerary is an optional enhancement if implementation time allows.

5.2 Wishlists & Saved Items

- **Saved Destinations** and Saved **Attractions** are separated for clarity.
- **Custom** folders can be created for different travel intentions.
- Each saved item **provides** View Detail, Add to Itinerary, and Remove from Saved.

6. Settings & Customization

6.1 Easy Account Setup

- **Phone login** includes phone input, verification code input, Send Code, Login, and validation messages.
- **Social login** buttons are grouped below phone login and use recognizable provider names/icons.

- Successful login transitions to Home; failed login explains the reason and next action.

6.2 Travel Preferences

- **Interest chips**: Adventure, Relaxation, Foodie, Culture, Nature, Shopping, Family, Photography.
- **Budget selector**: Low, Medium, High, Flexible.
- **Travel style selector**: Solo, Couple, Friends, Family, Short Trip, Long Trip.
- Preferences can be skipped and edited later, preventing onboarding from becoming a barrier.

6.3 System Settings

- **Language** and **currency** settings support common travel contexts.
- **Notification** settings cover itinerary reminders and saved trip updates.
- **Privacy** settings explain account, saved trip, and personal travel data visibility.
- **Appearance** settings allow users to switch between **light and dark mode** and customize the **theme color**. See "9.1 Visual & Color Coding" in detail.

7. Travel Guides & Preparation

7.1 Step-by-Step Planning Guides

Travel Guides support the core planning task instead of becoming a separate content platform.

- **Pre-trip checklist**: destination selection, date decision, attraction arrangement, packing reminder, and route check. These **reminders can be delivered** through system **pop-up prompts** and **notification reminders**.
- Contextual **guide cards appear when users create** the **first itinerary** or open the map for the first time.
- Local tips can be attached to destination details as short cards if data is available.

7.2 Preparation Widgets

Preparation widgets provide **quick access** to practical travel tasks. They are designed to reduce users' memory burden and help them check important information before departure. Packing checklist widget for manual items.

- **Packing checklist** widget: Users can manually add, edit, and check off packing items according to their travel needs.
- **Itinerary** review **reminder** widget: The app reminds users to review their itinerary before departure, including dates, saved attractions, routes, and activity times.
- **Offline** note **reminder**: When network access may be unstable, the app reminds users to save important notes, addresses, or itinerary information in advance.

8. Customer Support & Resources

8.1 Quick Access

User Support helps users recover from errors, learn the app quickly, and provide feedback. It appears in Profile, Help icons, empty states, and error prompts.

- **Help Center**: searchable FAQ covering login, search, itinerary editing, saved trips, map loading, and privacy.

- **Feedback** form: issue category, description field, optional screenshot placeholder, and Submit button.
- Error recovery: network failure, login failure, no search result, map loading failure, and save failure each provide a clear reason and next action.
- External airline/hotel/customer service links may be shown as supporting resources, not as core booking functions.

8.2 Community Support

- Destination tips can appear below destination details as short read-only cards.
- Reviews are read-only in the current scope unless implementation later supports submission.
- Community content should be clearly labeled as user-generated if implemented.

8.3 Onboarding and Contextual Help

The course material describes user support requirements such as availability, accuracy, consistency, robustness, flexibility, and unobtrusiveness. This app applies those requirements as follows.

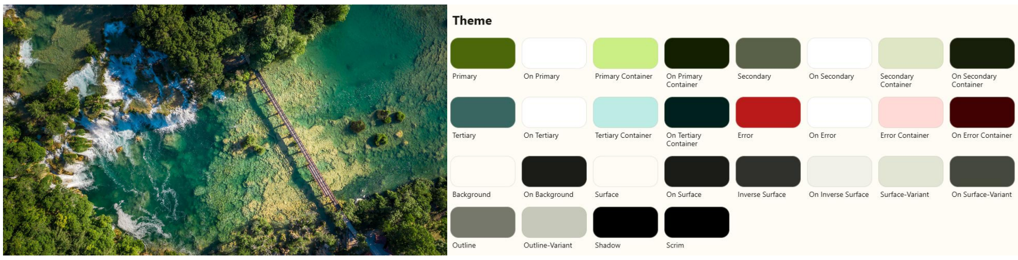
Support requirement	App implementation
Availability	Help is reachable from Profile, error prompts, and first-use tips without leaving the current task.
Accuracy and completeness	FAQ entries match actual app behavior and cover login, search, itinerary, map, saved trips, and privacy.
Consistency	Help text uses the same terminology as screen labels, buttons, and error messages.
Robustness	Help and error recovery still provide next actions when network or map data fails.
Flexibility	First-time users get tutorials; experienced users can dismiss tips and search directly.
Unobtrusiveness	Coach marks and assistants are user-controlled and do not interrupt core planning repeatedly.

9. Design Guidelines

9.1 Visual & Color Coding

The visual style follows **Material You / Material Design 3**. In this project, color is used to support hierarchy, interaction states, personalization, and accessibility without overpowering travel images, itinerary cards, or map content.

A key design decision is the use of Material You's **dynamic color mechanism**, also known as **Monet**. Monet generates a **complete Material 3 color scheme** from a **single source color**, such as the Android user's wallpaper or a user-selected theme color. The following example shows a Bing daily wallpaper and the Monet palette automatically generated from it..



Monet keeps screens visually consistent, supports light and dark modes, and reduces mismatched manual color choices.

The generated color scheme is reused across the interface:

- **Primary**: main actions such as Create Itinerary, Save, and Add to Itinerary.
- **Secondary/tertiary**: category chips, recommendation reasons, and supportive highlights.
- **Error**: validation errors, login failures, failed saves, and network unavailable states.
- **Surface**: cards, sheets, dialogs, grouped settings, and support content.

9.2 Typography

The typography also **follows** the **Material You** / **Material Design 3** type system. The app uses predefined typography roles instead of manually assigning unrelated font styles. **Roboto** or the system default font is used as the main typeface, and **Noto Sans** can be used when broader language support is needed.

Screen titles, **destination names**, **itinerary details**, **buttons**, **chips**, and **navigation labels follow** Material 3 typography **roles** such as Title Large, Title Medium, Body Medium, and Label Large. The design also support system font scaling and variable fonts without text overlap, clipped labels, or blocked controls.

9.3 Material 3 Component Mapping *(Newly Introduced)*

Material You updates and refines the component appearance of Material Design. In this project, we **continue to use** the **mature component** solutions **provided** by **Material You**. This helps the interface remain consistent, modern, and easy to maintain across navigation, search, actions, editing, feedback, settings, and support functions.

UI need	Material 3 component	Usage in this app
Primary navigation	Navigation bar	Home, Explore, Itinerary, Saved, Profile
Search and discovery	Search bar, filter chips	Destination search and category filtering
Recommendation display	Filled/elevated/outlined cards	Recommended destinations and saved trip cards
Main actions	Filled button, FAB	Create itinerary, add activity, save plan
Compact editing	Modal bottom sheet	Add/edit itinerary item and marker details
Status and confirmation	Snackbar, dialog	Saved, deleted, login failed, network unavailable
Settings	Lists, switches, chips	Language, notification, budget, travel preference
Support	Searchable list, FAQ cards, text fields	Help Center and feedback form

9.4 Accessibility Review

The accessibility review uses **WCAG** (Web Content Accessibility Guidelines) 2.1 AA-oriented checks from the requested accessibility-review skill and adapts them to a mobile prototype context.

Area	Requirement	Design action
Contrast	Normal text target >= 4.5:1; large text/UI graphics target >= 3:1.	Use tested Material color roles; do not rely on wallpaper color if contrast fails.
Touch targets	Common interactive targets should be at least 44x44 CSS pixels.	Apply to chips, map pins, card actions, bottom navigation, and edit/delete controls.
Text scaling	Readable at larger font settings without clipped text.	Use flexible card height and avoid fixed small labels in critical flows.
Keyboard/focus	Logical focus order and visible focus indicator where supported.	Order navigation, search, cards, bottom sheet actions, and dialogs predictably.
Screen reader	Components expose name, role, and state.	Provide labels for icons, map controls, chips, switches, and image cards.
Error messages	Errors are identified in text, not color alone.	Show inline text plus status icon/field state; preserve entered data after failure.
Non-text content	Meaningful images and diagrams need descriptive alternatives.	Add captions/alt descriptions for figures and destination images.

9.5 Performance

Performance is important for both usability and user experience. Several optimizations are applied to reduce waiting time and improve perceived performance.

- Use **lazy loading** for destination images and avoid blocking the first screen with large image assets.
- Show **skeleton placeholders** or progress indicators for recommendations and map data.
- Keep map marker **rendering lightweight** and cluster/filter dense markers.
- Core interactions such as switching tabs, opening cards, and editing itinerary items should feel immediate.

9.6 Design Critique Checklist *(Newly Introduced)*

After the main design is completed, we will answer the following checklist questions to further optimize the interface and code logic. These questions help us check whether the app supports the core travel-planning workflow clearly, consistently, and reliably.

Checklist question	Pass condition
What draws the eye first?	Search, current trip, and Create Itinerary are more prominent than secondary content.
Can the user accomplish the main goal?	A new user can create and save a two-day itinerary without instructor explanation.
Is the navigation intuitive?	The five bottom tabs match the user's mental model and use stable labels.
Are visual elements consistent?	Cards, chips, buttons, sheets, and dialogs reuse the same spacing, shape, and color roles.
Are errors recoverable?	Login, search, map, and save failures provide readable cause and next action.
Is accessibility considered?	Contrast, target size, text scaling, labels, and non-color status indicators are specified.

10. Evaluation Design (Week 16 Requirement)

The evaluation design **combines questionnaire feedback**, task-based usability **testing**, expert **review**, and **accessibility checks**. This is stronger than a questionnaire-only plan because the course slides emphasize that evaluation should test usability and functionality, identify specific problems, and occur throughout the design life cycle.

From Section 10.5 to the end of Section 10, the newly introduced content explains how we test and evaluate the app. It defines usability metrics, test procedures, reporting methods, ethical considerations, and success criteria to guide later interface and code improvements.

10.1 Evaluation Objectives

- **Validate** whether the Home Screen communicates the product purpose and main actions within a short first impression.
- **Measure** whether users can create, edit, and save a basic itinerary without detailed instruction.
- Check whether search, filters, and recommendation cards help users find relevant destinations efficiently.
- **Evaluate** whether map markers and bottom sheet actions support spatial understanding and itinerary planning.

- **Verify** that support resources, error messages, and accessibility decisions help users recover from problems.

10.2 Participant Recruitment

- **Target** participants: university students, young travelers, weekend travelers, independent travelers, and ordinary mobile users.
- **Questionnaire** sample: at least 10 volunteers, as required by the template. **See 10.3.**
- **Usability-test** sample: 5-8 participants if schedule allows; otherwise at least 3 pilot participants plus questionnaire data. **See 10.5.** **(OPTIONAL)**
- Other Aspects.

10.3 Questionnaire

Use a five-point scale: 1 = Strongly Disagree, 5 = Strongly Agree. Add two open-ended questions at the end.

No.	Question	Evaluation focus
1	I can understand the main functions from the Home Screen quickly.	Home clarity
2	The bottom navigation makes it easy to move between Home, Explore, Itinerary, Saved, and Profile.	Navigation
3	The search bar and category chips help me find destinations efficiently.	Search/discovery
4	The recommendation cards are clear and useful for travel planning.	Recommendation UI
5	The reason shown on a recommendation card helps me understand why it appears.	Recommendation transparency
6	Creating a new itinerary is simple and intuitive.	Itinerary creation
7	Editing, deleting, and reordering itinerary items are easy to understand.	Itinerary management
8	The map markers and bottom sheet help me understand destination locations.	Map interaction
9	Saved Trips and Wishlists are organized clearly.	Saved content
10	Login and account setup are simple and provide clear feedback.	Authentication
11	The Help Center, FAQ, and error prompts can help me solve problems.	User support
12	The visual style is clean, modern, and consistent.	Visual design
13	Text, buttons, chips, and map actions are easy to read and tap on a phone.	Accessibility
14	The app gives enough feedback when something is saved, loading, invalid, or unavailable.	System feedback
15	Overall, I would be willing to use this app for travel planning.	Satisfaction

Open-ended Q1: Which part of the app felt most confusing or unnecessary?
Open-ended Q2: What is one feature or screen change that would make this app more useful for your travel planning?

10.4 Reporting and Data Processing

The goal is to combine quantitative results and qualitative observations, identify key usability problems, and convert them into actionable design improvements.

Quantitative data will be summarized through mean scores for each questionnaire item, task completion rate, task time range, and the number of assists or errors. **Qualitative data** will be analyzed through affinity mapping, where user comments and observation notes are grouped into themes such as navigation, map interaction, itinerary editing, support, and accessibility.

Identified issues will then be classified by severity: Critical issues block task completion, Major issues cause delay or repeated errors, and Minor issues cause confusion but have an obvious workaround.

The **final report** will present the findings with clear visualizations and improvement recommendations. Bar charts can be used to show mean questionnaire scores, stacked charts can show task success, partial success, and failure, and pie charts can summarize overall satisfaction if needed. Each improvement recommendation should include the affected screen, supporting evidence, issue severity, and the proposed UI change.

10.5 Usability Tasks

#Task	Scenario	Expected success
T1	Log in with phone verification or choose social login path.	User reaches Home or understands login feedback.
T2	Find a weekend destination using search and category chips.	User opens a relevant destination detail page.
T3	Create a two-day itinerary and add one activity.	User creates a trip with at least one saved itinerary item.
T4	Open the map, select a marker, and add it to the itinerary.	User understands marker details and completes Add to Itinerary.
T5	Edit, reorder, or delete an itinerary item.	User completes the edit and understands saved/error feedback.
T6	Find the saved trip and locate Help Center or FAQ.	User reopens the trip and finds support without assistance.

10.6 Metrics and Success Criteria

Following the usability engineering slides, metrics are organized around effectiveness, efficiency, satisfaction, learnability, and error tolerance.

Metric category	Measure	Planned success criterion
Effectiveness	Task completion rate for T1-T6	At least 80% of participants complete T2, T3, and T6 without evaluator intervention.
Efficiency	Time to complete major tasks	Most participants complete the core search-to-itinerary flow within a reasonable course-test time window.
Error tolerance	Number and severity of mistakes; recovery success	Critical errors are rare, and users can recover from delete, invalid input, and network/map failure states.
Learnability	First-task hesitation and requests for help	Users can explain what Home, Explore, Itinerary, Saved, and Profile mean after short exposure.
Satisfaction	Mean Likert score and open comments	Average rating for navigation, itinerary creation, map clarity, and overall satisfaction is >= 4.0/5 if possible.
Accessibility	Contrast, target size, text scaling, labels, and error text	No known critical accessibility issue remains before final presentation.

10.7 Test Procedure

1. Welcome the participant, explain the purpose, and confirm consent. The test evaluates the design, not the participant.
2. Collect short background information: travel frequency, smartphone familiarity, and previous travel-planning habits.
3. Show the prototype or UI screens without explaining the intended path.
4. Ask the participant to complete tasks T1-T6 while thinking aloud.
5. Record completion, time, errors, assistance, hesitation points, and notable comments.
6. Ask the questionnaire and short debrief questions.
7. Summarize issues by severity and connect each issue to a screen, requirement, or evaluation task.

10.8 Ethics, Limitations, and Validity

- Participants should be told that participation is voluntary and that only design feedback is being evaluated.
- No sensitive travel or account data should be collected during the test.
- A course sample of 10+ volunteers is useful for formative feedback but should not be treated as a statistically representative market study.
- Prototype limitations should be reported clearly, especially if some flows are simulated rather than fully implemented.

10.9 Requirement-to-UI and Evaluation Traceability

This section connects the original Project requirement of Stage 1, modules with the corresponding UI screens and evaluation coverage. It helps verify that each major requirement is represented in the interface design and can be checked through a task, questionnaire item, heuristic review, or accessibility review.

Project1 requirement module	UI screen / component	Evaluation coverage
Registration and Login	Login/Register screen, Profile session state	Task T1; questionnaire item 10; error feedback review
Destination Browsing and Search	Home search, Explore search, filters, destination cards	Task T2; items 1, 2, 3; search no-result observation
Destination Detail Viewing	Destination detail page, Save/Add/Open Map actions	Task T2/T4; recommendation and action clarity review
Personalized Recommendation	Recommended for You cards and explanation text	Items 4 and 5; first-impression critique
Interactive Map Viewing	Map screen, markers, bottom sheet, marker filters	Task T4; item 8; accessibility target-size review
Itinerary Planning and Editing	Timeline, bottom sheet editor, Save, Edit, Delete, Move	Tasks T3 and T5; items 6 and 7; error recovery review
Saved Trip Management	Saved tab, wishlist folders, past trips	Task T6; item 9
Fault Handling	Inline validation, dialogs, snackbar, retry/empty states	Items 11 and 14; heuristic/accessibility review
Operating Environment	Mobile-first Material 3 layout, text scaling, performance states	Item 13; accessibility/performance review

References

1. Software Requirements Specification from Project, Stage 1 (Week 5)
2. Google Material Design 3 / Material You official guidelines: https://m3.material.io/
3. Android Mobile App Developer Tools - Android Developers. https://developer.android.com/

Appendix A. Other UML Diagrams.

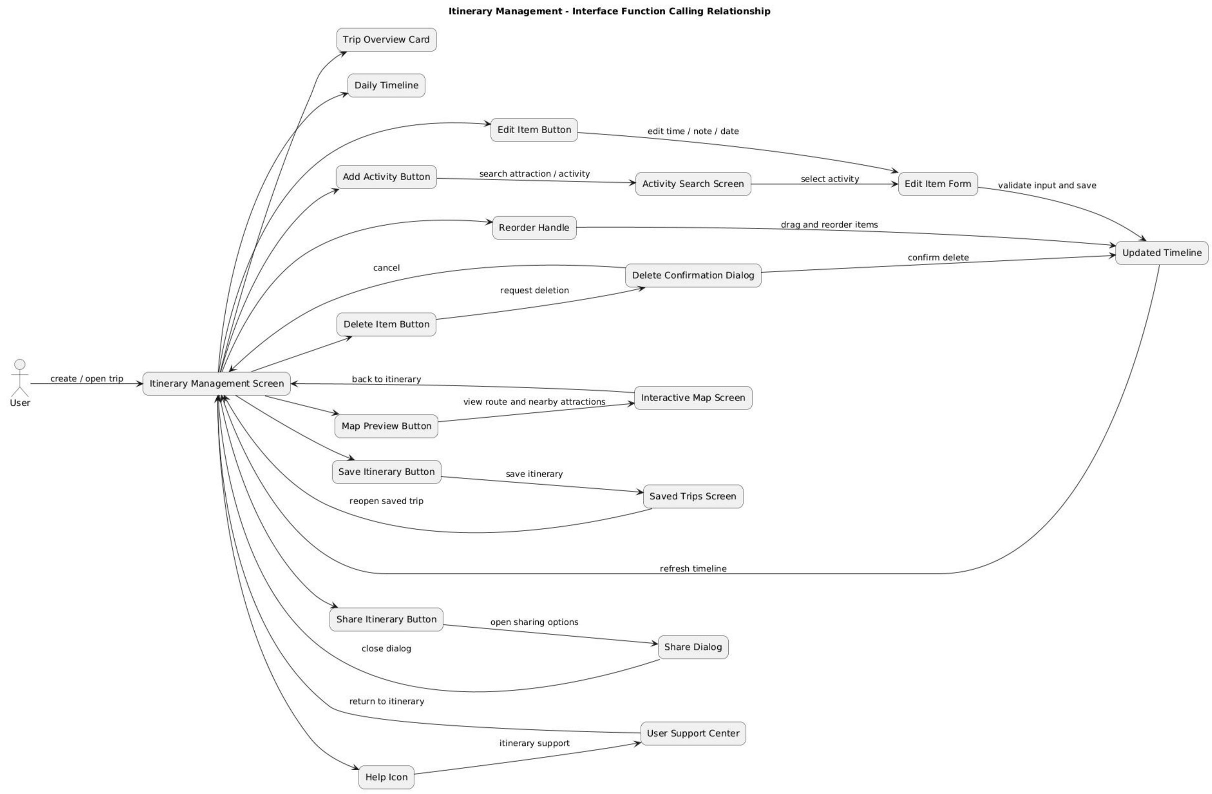


Figure A1. Itinerary management interface-function calling relationship.

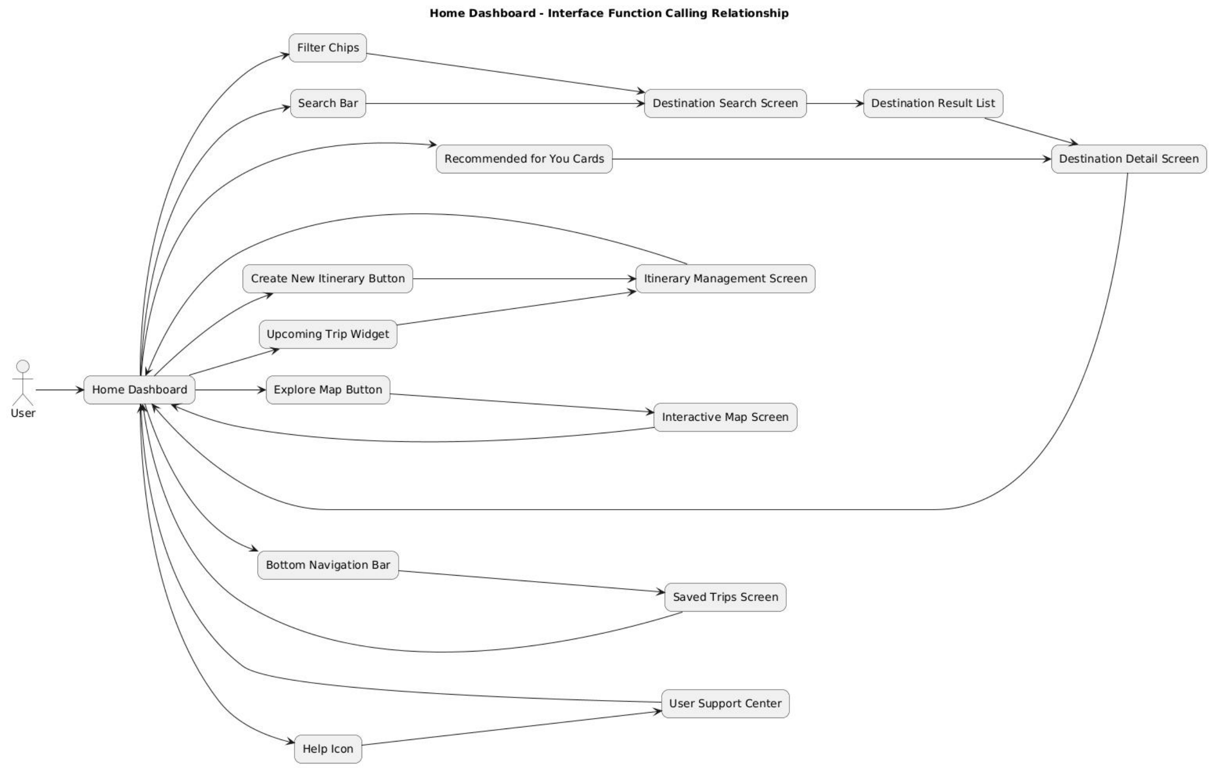


Figure A2. Home dashboard interface-function calling relationship.

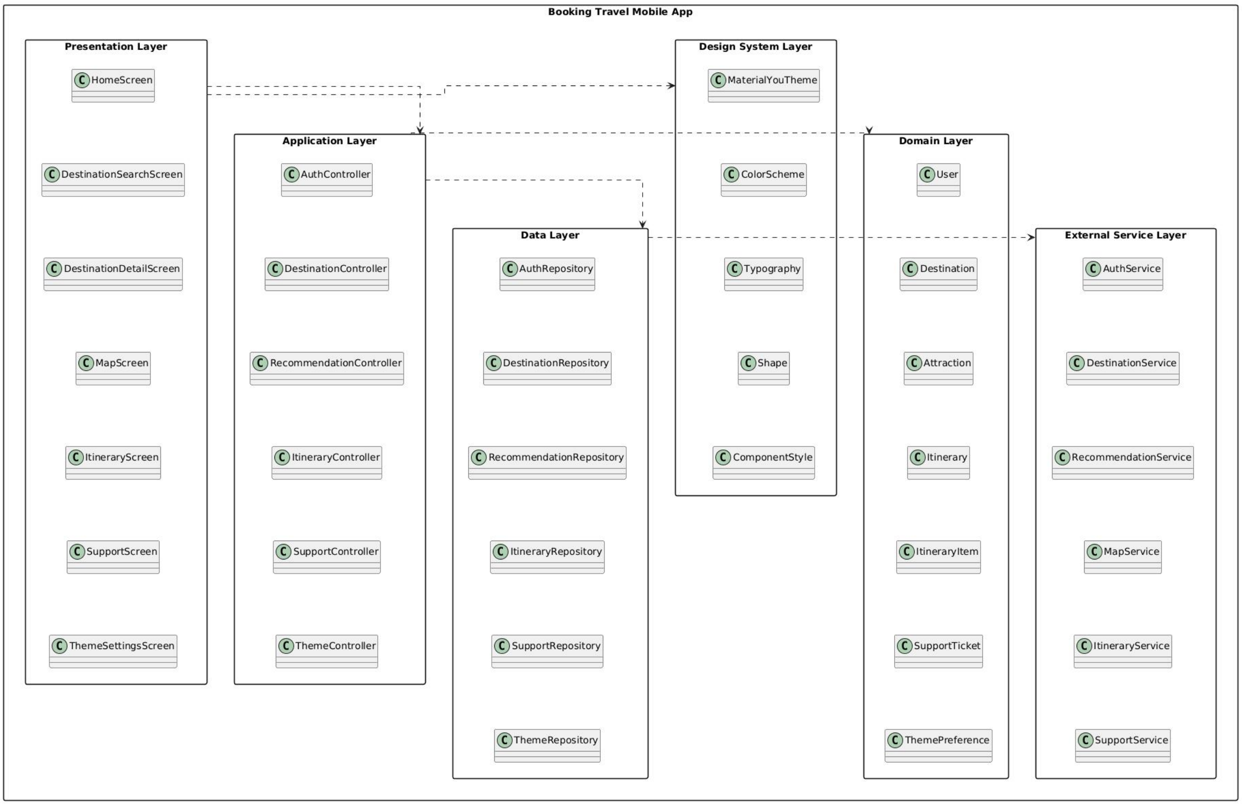


Figure A3. Functional layer structure for UI, service, storage, and external services.

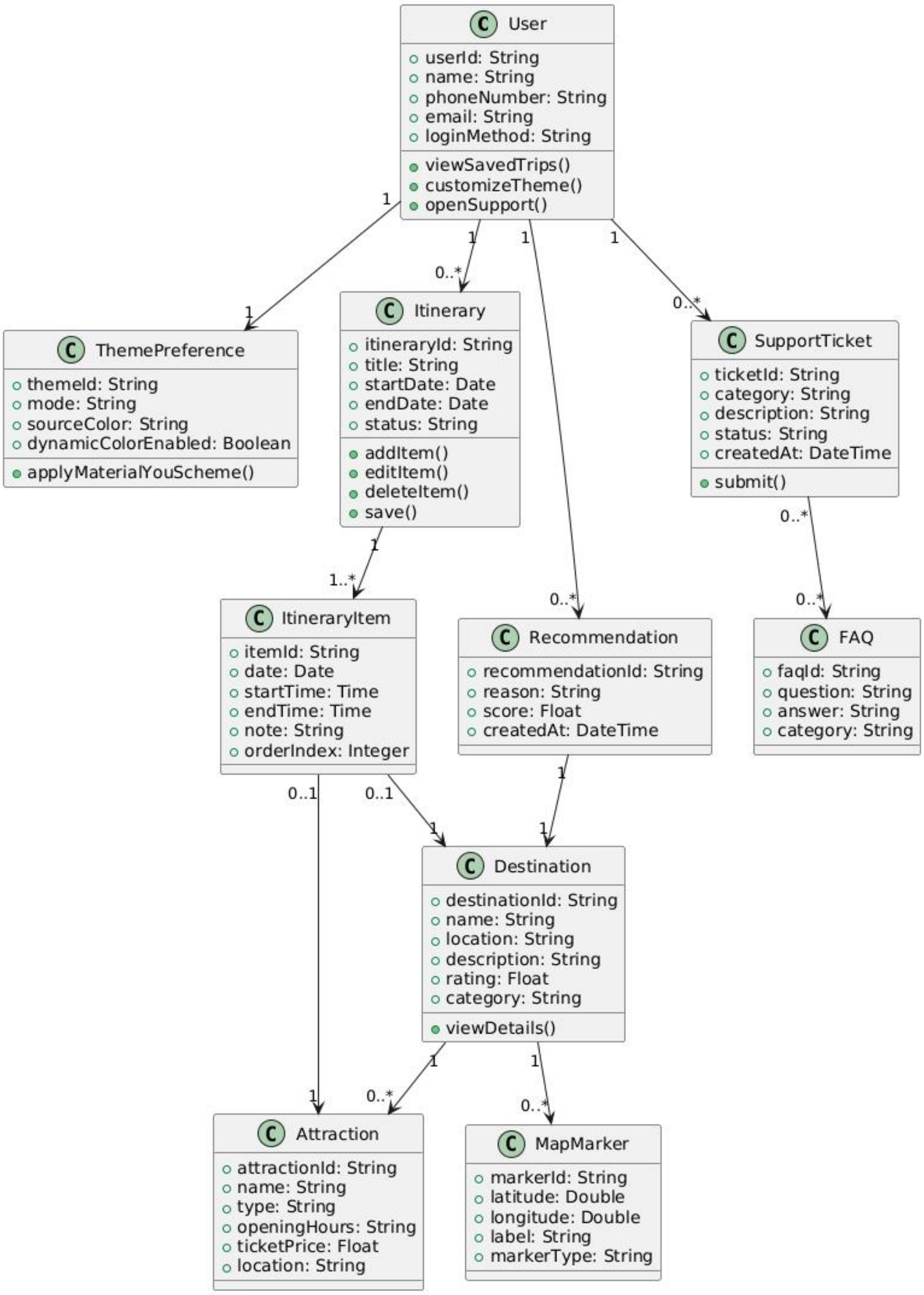


Figure A4. Core data/entity relationship sketch.

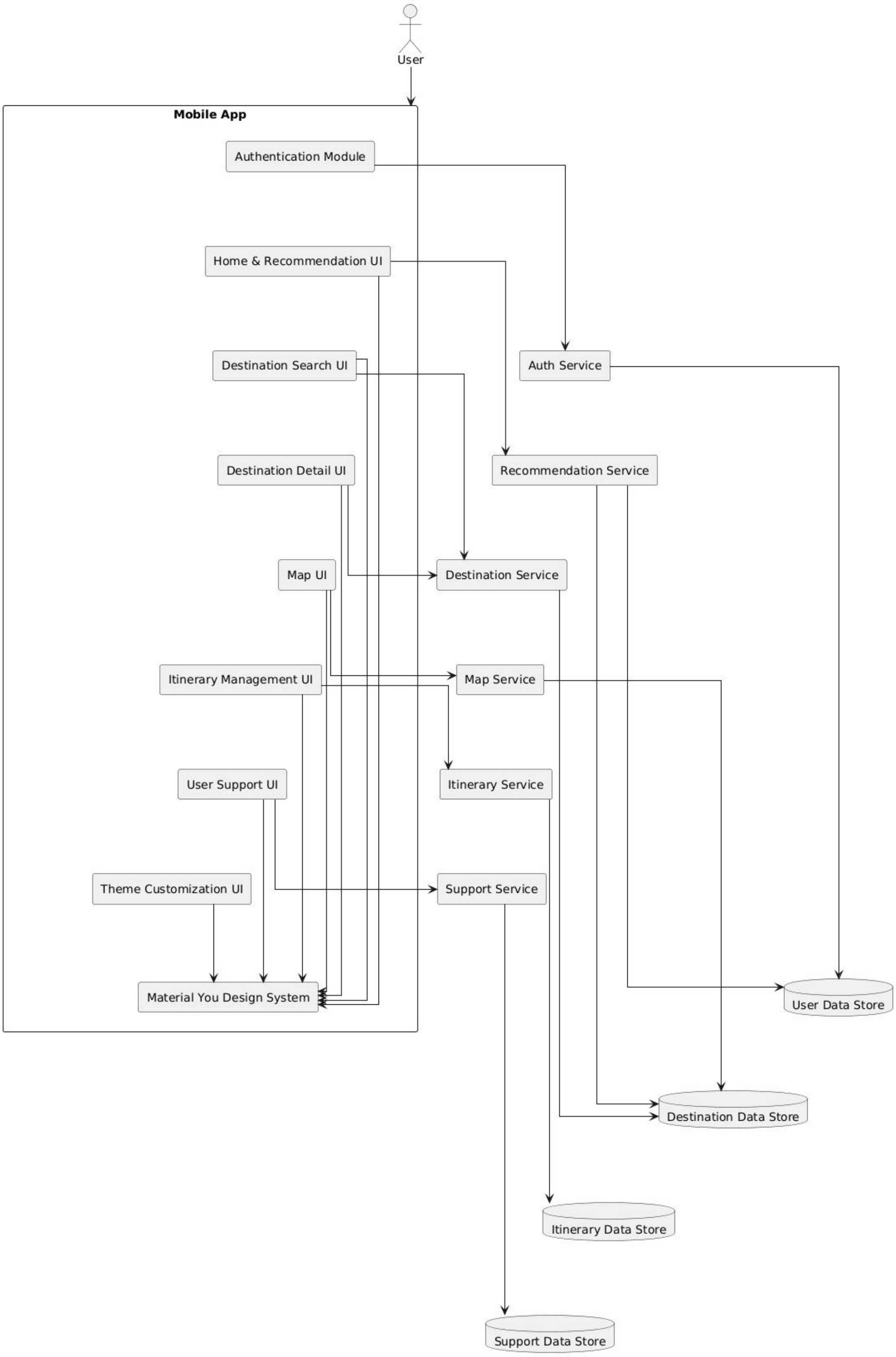


Figure A5. Mobile app and service interaction overview.

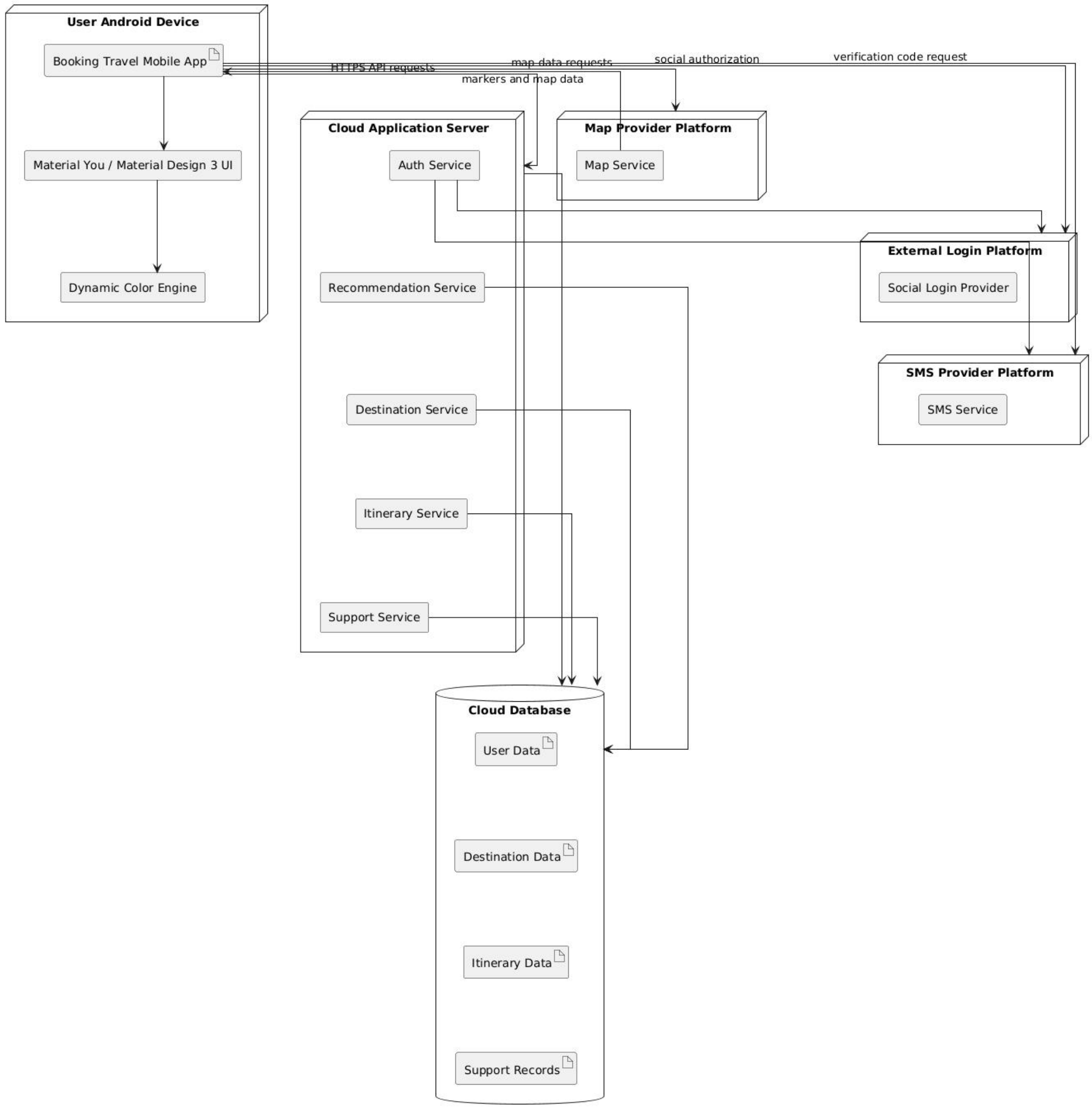


Figure A6. Deployment and external platform relationship.